



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

PROJECT 3:

GROUP 3

SECV2113-01 HUMAN COMPUTER INTERACTION

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1.0 Storyboards

1.1 Task 1: Design an integrated healthcare record management system



Scene 1:

The user struggles with disorganized health records. The user is surrounded by a messy pile of papers, looking confused.

Scene 2:

The user sees an advertisement on their phone about an app that organizes health records.

Scene 3:

The user downloads the health record management app. The user clicks "Download" on the app store.

Scene 4:

The user uploads health records into the app. The user takes a photo of a medical document with their phone.

Scene 5:

The app provides a health summary. The app screen shows simple graphs and icons summarizing the user's health data.

Scene 6:

The app gives personalized health tips. The user sees a list of personalised health tips with recommended healthy food and some advice.

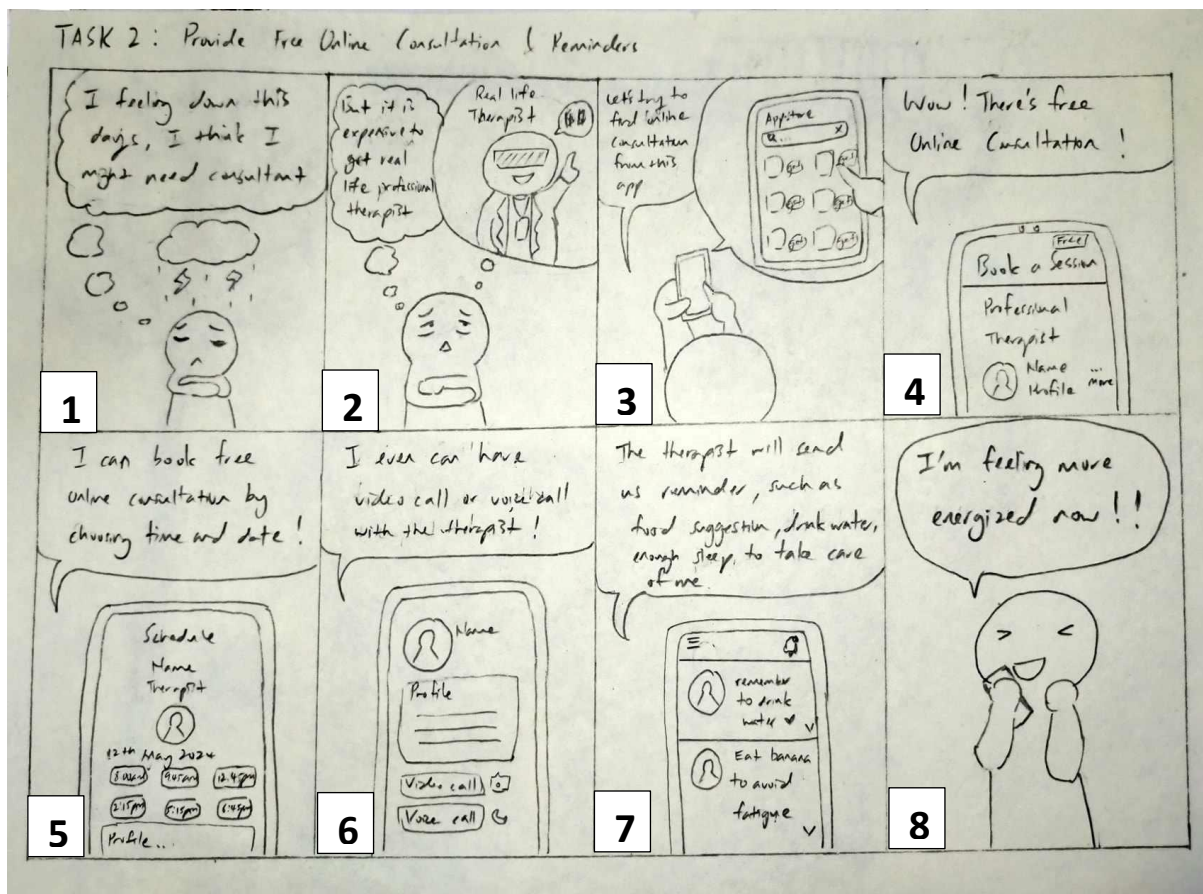
Scene 7:

The user book a medical appointment and sets reminders for appointments for it. medical appointment and sets reminders for appointments and exercise.

Scene 8:

The user feels in control of their health.

1.2 Task 2: Provide Free Online Consultation & Reminders



Scene 1:

The user is feeling down around this day, and thinks that he might need consultant.

Scene 2:

The user finds out that seeing a personal therapist in real life is expensive.

Scene 3:

The user decides to download an app to get an online consultant.

Scene 4:

The user discovers that there is an app offers a free online consulting service.

Scene 5:

The user can book a free online consultant by choosing a time and date.

Scene 6:

The user can have a video call or voice call with a professional therapist.

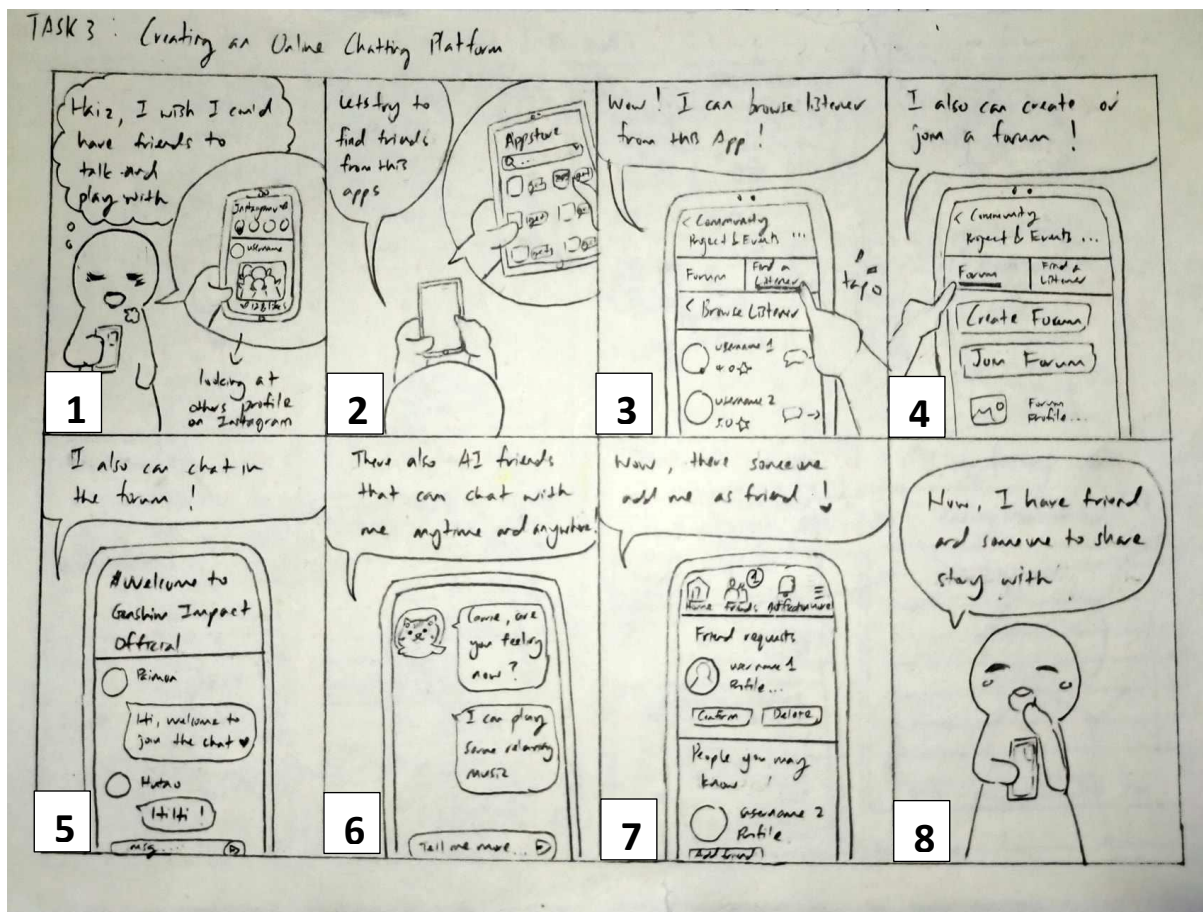
Scene 7:

After the consultant, the user will receive reminders from his personal therapist, such as food suggestions, to drink water, and to get enough sleep.

Scene 8:

User feels more energized now.

1.3 Task 3: Creating a Online Chatting Platform



Scene 1:

The user feels lonely and wishes he had a close friend to talk and play with.

Scene 2:

The user decides to try finding a friend through an app he finds in app store.

Scene 3:

The user discovers the app allows him to browse listeners and choose anyone to talk to.

Scene 4:

The user finds out he can also create or joins a forum that interest him.

Scene 5:

The user can chat with other users and share his thoughts or stories in the forum.

Scene 6:

The user discovers that the app has also AI friend that he can chat with anytime and anywhere.

Scene 7:

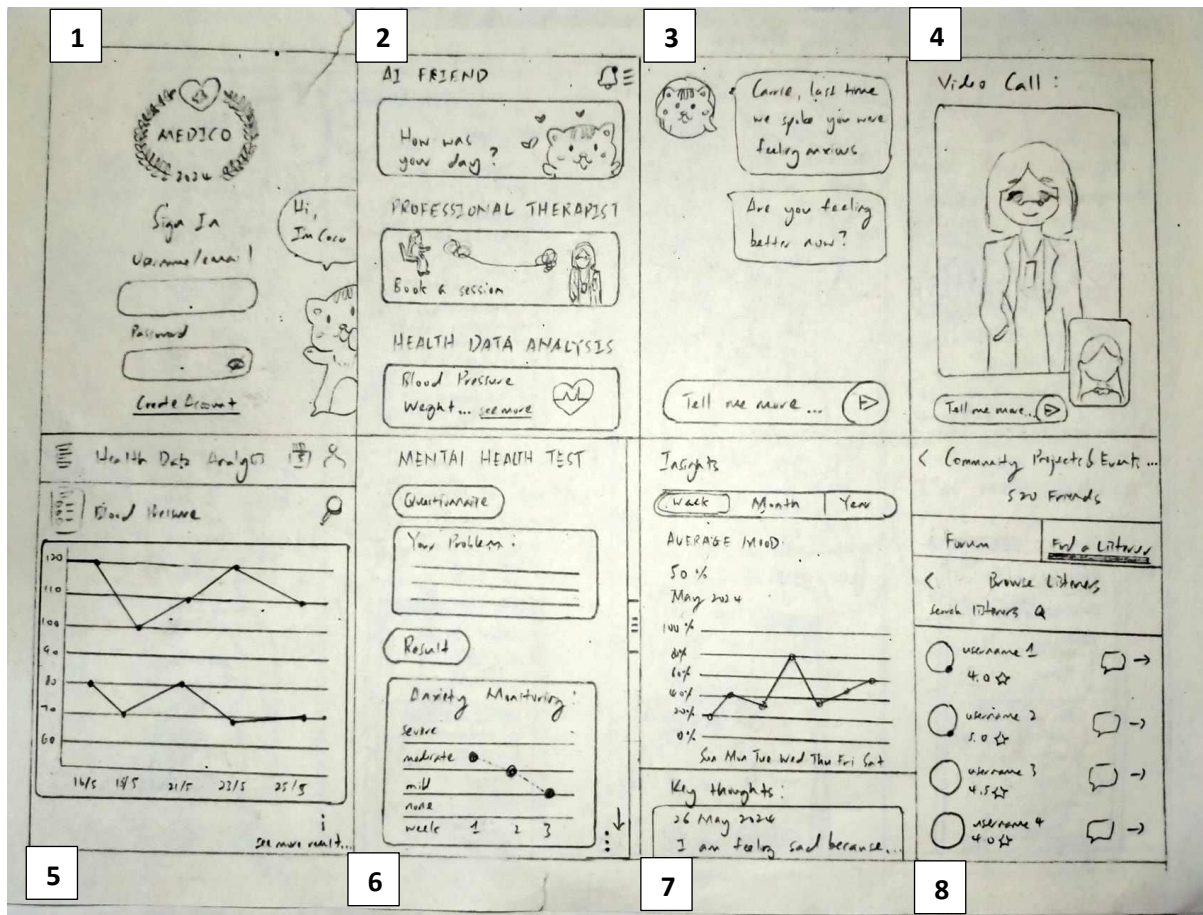
The user is pleasantly surprised that someone is willing to be friend with him.

Scene 8:

The user is happy to have a friend now to share stories and play with.

2.0 Alternative Design

2.1 CARRIE TEO HUI XIN



Scene 1:

This page shows that default start-up page of the application which is the sign-in page. On this page, there are two main parts, which are the sign in, and create account, where sign-in is for the existing user to log in, create account is for the new user registration.

Scene 2:

This page is the main page or the home page of the application which contains three main functions which will bring the user to the respective page such as AI friend chatting platform, free online consultant service, and health data analysis. There is also reminder sign on top of the screen to notify the user. Besides the reminder sign, there is also a sign where user can click to see their profile.

Scene 3:

This page is where the user can chat with the AI friends. The main point of this page is that users can chat with an AI friend is to get support anytime and anywhere, especially for users who might feel lonely or in need of someone to talk to. This feature ensures that users always have access to a friendly, responsive conversation partner, helping to alleviate feelings of isolation and enhance their emotional well-being.

Scene 4:

This page is the page where the user can make video call or voice call with the professional therapist after booking a free online consultant service. The main point of this page is to facilitate seamless communication between the user and a professional therapist. Users can initiate a call at their scheduled appointment time, ensuring they receive timely and personalised support. The interface is user-friendly, providing clear options to start or join the call. This page ensures that users have a straightforward and efficient way to access professional mental health support, enhancing their overall experience and well-being.

Scene 5:

This is the page where the application will display all result of the health data analysis. The main point of this page is to provide users with comprehensive health data analysis, with a key focus on monitoring blood pressure. This page displays an in-depth analysis of all health data, offering insights into various health metrics, such as blood pressure to help us stay informed about our health. Users can view trends and patterns in their blood pressure readings over time, set personalised goals, and receive alerts if their readings fall outside the normal range.

Scene 6:

This page enables users to complete a questionnaire and list their problems, allowing for an analysis of their mental health status. The main point of is page is to provide users with a comprehensive questionnaire designed to assess various aspects of their mental health. Users can list out their problems, feelings, and experiences in detail. The information gathered through this questionnaire helps in analysing the user's mental health status, identifying potential issues, and tailoring personalised support and recommendations. This page is crucial for providing a baseline understanding of the user's mental health, which is essential for effective diagnosis and treatment planning.

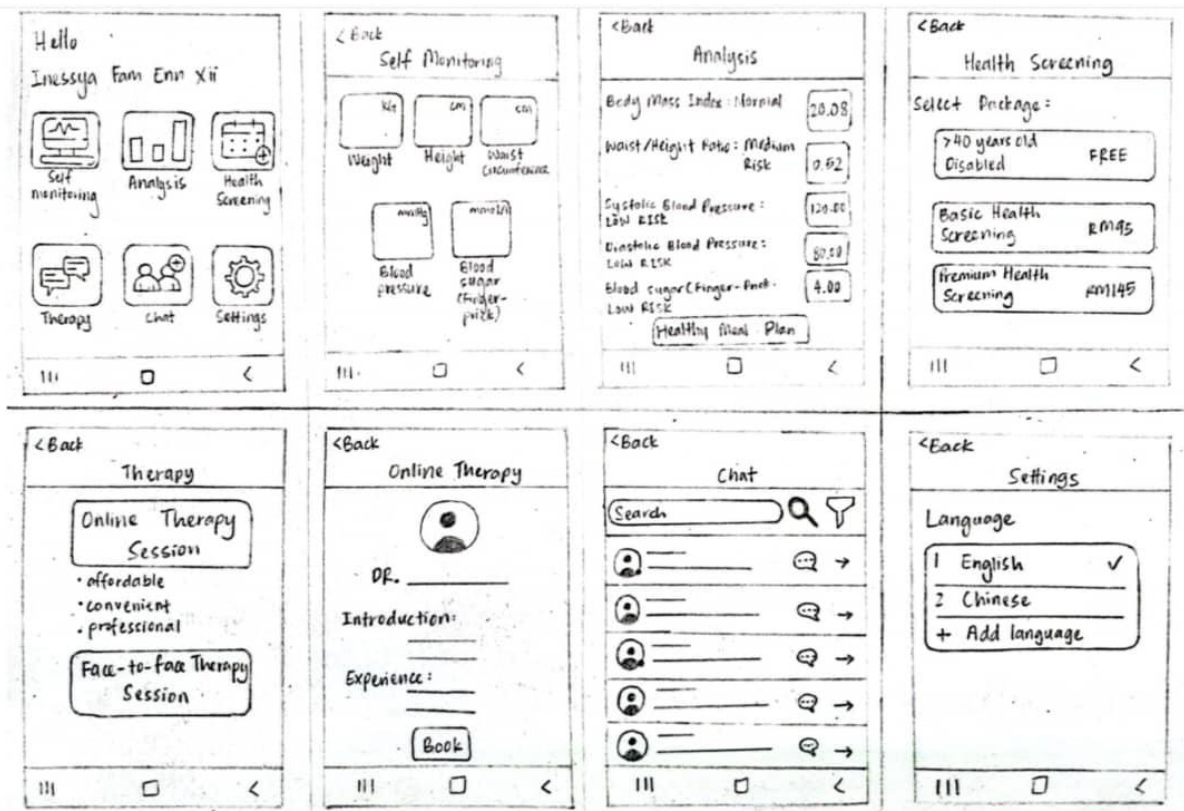
Scene 7:

This page provides users with insights into their average mood over a week, month, or year, based on the daily stories they share in the key thoughts section. The main point of this page is to present users with detailed insights into their average mood over different time periods—weekly, monthly, or yearly—derived from the daily stories and key thoughts they record. By tracking and analysing these entries, the app generates visual representations and summaries of mood trends, helping users understand their emotional patterns and fluctuations over time. The interface includes charts and graphs that highlight significant changes and trends, offering a clear and comprehensive view of the user's emotional well-being. This feature empowers users to gain a deeper understanding of their mental health, identify potential triggers, and make informed decisions to enhance their overall emotional wellness.

Scene 8:

This page to enable users to find and connect with a listener to chat with, share stories, and make friends. The main point of this page is to allow users to search for and connect with listeners who are available for chat. Users can share their stories, discuss their experiences, and build friendships through these interactions. The interface is designed to make it easy to browse through potential listeners, view their profiles, and initiate conversations. This feature fosters a supportive community environment where users can find companionship, emotional support, and meaningful connections, enhancing their social well-being and reducing feelings of loneliness.

2.2 INESSYA FAM ENN XII



Scene 1:

This page shows the home page of the app after the user has logged into their account. The screen of the home page shows some functions provided in this application. Users can click on the icon to be directed to the services function. The first icon is the self-monitoring button, which is for the user to enter their health data. The second icon is for the analysis of the user's health data. The third icon is for the user to book an appointment for health screening. The fourth icon is for the user to book a therapy session. The fifth icon is for the user to chat with other users who are using this app to find listeners and make friends. The sixth icon is the settings button.

Scene 2:

When the user clicked the self-monitoring icon, they will be directed to this page. This page allows users to record their health data such as Weight, Height, Waist circumference, Blood pressure and Blood sugar (Finger-prick).

Scene 3:

When the user clicked the analysis icon at the home page, they will be directed to this page. This page shows the results analysed from the user's health data input. The result will show the Body Mass Index, Waist/Height Ratio, Systolic blood pressure, Diastolic blood pressure, and Blood sugar (Finger-prick). The boxes with the health data inside will appear in three different colours which are red, orange and green to allow user to understand whether they have high or low risk, where green stands for low risk and red stands for high risk. There will be a "Healthy Meal Plan" button at the bottom of the page. When user clicks the button, healthy meal plans will be suggested to the user based on their current health condition.

Scene 4:

When the user clicked the health screening button at the home page, they will be directed to this page. This page shows three different health screening packages for the user to select based on their preference. For elderly users and users with disabilities, they can benefit from choosing the free health screening package to ensure they have easy access to health screening regularly. The other two packages are basic health screening, RM95 and premium health screening, RM145.

Screen 5:

When the user clicked the therapy button at the home page, they will be directed to this page. This page shows two choices of therapy for the user, which is online therapy and face-to-face therapy with professional therapists.

Screen 6:

When the user clicked the online therapy button at the therapy page, they will be directed to this page. This page shows the therapist's profile picture, name, short introduction and their experience in this field. Then, if the user decided to choose the therapist, they can just click the "book" button below the experience section.

Screen 7:

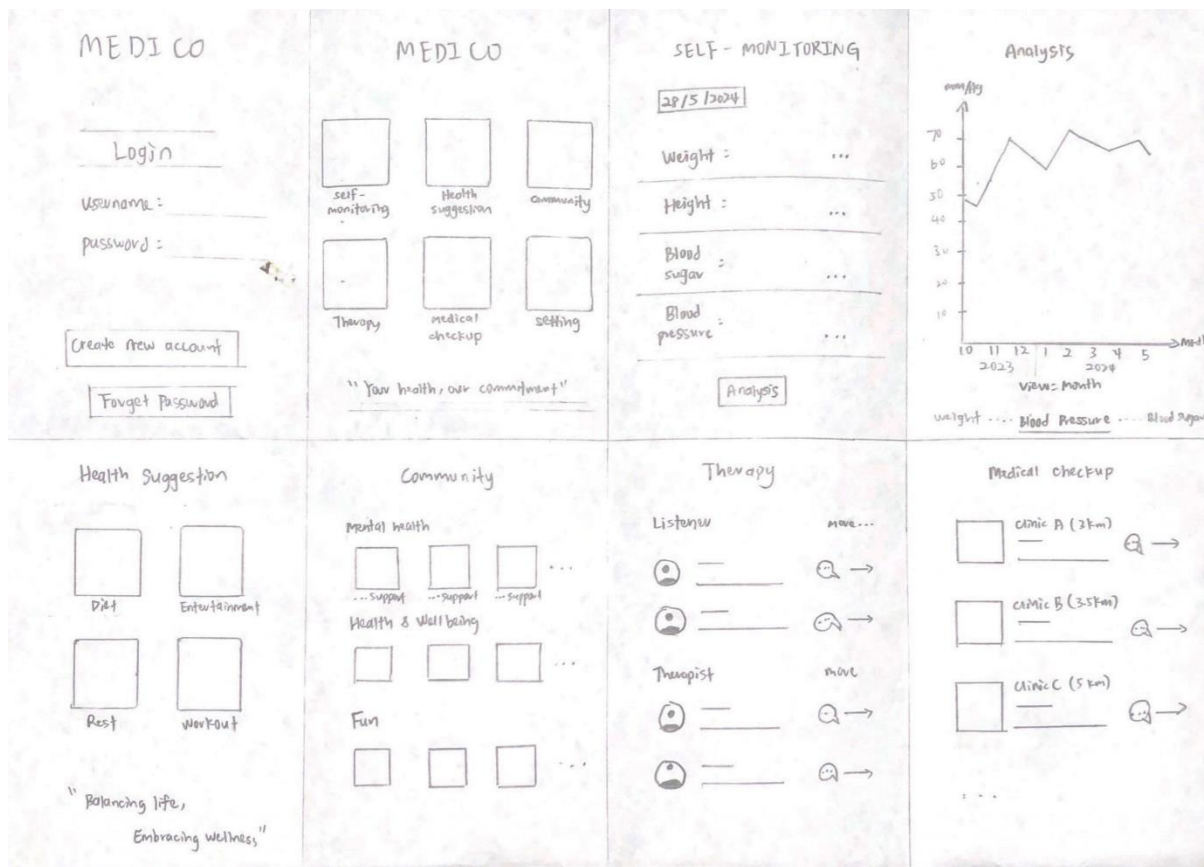
When the user clicked the chat button at the home page, they will be directed to this page. This page shows other users in this app that can be chosen as listener by the user. The user can filter the other users shown in this page to filter them based on countries to eliminate the problem of language barrier. The user can also search for other users' name in the search bar. When the

user has decided which user to chat with, they can just click the chat icon on the left side of every user's account.

Screen 8:

When the user clicked the settings button at the home page, they will be directed to this page. This page shows language settings of this app. The user is free to choose which language they want to use in this app so they can understand how to use this app better.

2.3 LEE YIN SHEN



Scene 1:

First of all, user need to login the app by entering their username and password. For new user, they need to click the create new account button to create their account. They should add their email for verification. Also, for those who forgot their password, they can press the forgot password button then the system will send a verification code to their registered email.

Scene 2:

After the user login the app, they can see a very user-friendly and straightforward interface, which display all of the icons of functions they can access, which are self-monitoring, health suggestion, community, therapy and medical checkup. For setting, the user can add their password, verified email, phone number, gender and other personal detail. Moreover, our app provide three main languages which are English, Chinese and Malay that enable the user to select.

Scene 3:

For the first function “SELF-MONITORING”, the user can input their personal health data like weight, height, blood pressure and blood sugar level. The user can input their health data based on the date, and if they miss to input the previous health data, they can add it by pressing the date above. Also, they can also check their previous health data at this function. This enable the user to record their daily health data for further analysing. Then, there are a “Analysis” button at the button, which enable the user to look at their health data in graph representation.

Scene 4:

After they press the “Analysis” button from scene 3, the users are able to see the analysis of their health data in graph representation. We provide three type of view which include yearly view, monthly view and weekly view of their health data to enable the users to look at their changes in their health. They can change the view type easily by click the “View:” button. Furthermore, they can select which health data they want to view by dragging the health data type at the bottom.

Scene 5:

For health suggestion function, the users are able to explore some tips focused on diet, entertainment, sleeping and workout by clicking the buttons of each category. If they click the diet icon, the users can view various recipes on healthy and delicious foods which provided by professional nutritionist. If they click the entertainment icon, the users are able to see how can obtain entertainment in a healthier way. For example they can watch some relaxing video, listen to the music and ext. Besides, if they press rest icon, they can view the tips on how to rest effectively like how many hours to sleep based on their age, and the correct posture to sleep. Lastly, for the workout category, the users can access various of workout plan suitable for different age and different physical level.

Scene 6:

For community function, the users can access various community based on their situation. For example a user got mental health problem, they can choose mental health category and they can enter the mental health support community. Inside the community, the users can exchange their ideas, share their problem and discuss together. This can be a platform for the users to share their life, grumbling and meet new friends.

Scene 7:

The users can also access the therapy function. For this function, the user can either choose find a listener or therapist based on their situation. They can view the personal information of those therapist and listener to select who they want to chat to. They can add the “-->” button to view more detail of the listener or therapist or they can directly click the chat icon to start chatting with them. They can also access more listener and therapist by clicking the more button.

Scene 8:

The last function will be the medical checkup function. In this function, the users are able to book medical checkup appointments easily and straightforward. After they click the medical checkup icon from the main interface, they will see a list of clinics nearby their current location. After they select the clinic they want to access, they can click the “-->” button for further detail. For the detail, the users can view the medical checkup packages provided in that clinic, and they will see a requirements list. If they match the requirement, they can directly contact the clinic and make an appointment.

2.4 ELIJAH SHE YU SHENG



Scene 1

In the first scene, users encounter the login interface for the MEDICO app. Here, they need to enter their username and password to sign in. For new users, there is an option to create a new account by clicking the "Login /Sign in" button, which requires email verification. Additionally, if users forget their password, they can click the "Forgot password" button. The system will then send a verification code to their registered email to help them reset their password. A small heart icon provides help if users need assistance with the login process, and a back button is available for navigation.

Scene 2

After logging in, users are presented with three main choices: find friends, view health details, or find a therapist. This interface is designed to be straightforward, with large, easily recognizable icons representing each choice. Users can select "Friends" to connect with others, "Health" to manage their personal health information, or "Therapy" to seek professional help. This setup allows users to quickly access the primary functions of the app based on their immediate needs.

Scene 3

The third scene expands on the communication options available within the app. Users can choose from three modes: one-on-one chatting, group chats, or community forums. This interface also includes a search bar at the top, enabling users to find specific people, groups, or communities. Additionally, there is a call button for initiating voice calls with contacts. This layout ensures users have multiple ways to communicate and connect with others within the app.

Scene 4

In the fourth scene, users are introduced to the "Self-Health Box" interface, featuring four cute buttons. The first button, represented by a clipboard icon, is for health reports and analysis. The second button, symbolized by a stethoscope, is for body checkups. The third button, depicted with a computer icon, leads to self-monitoring, where users can log their weight, height, blood sugar, and other health metrics. The final button, marked with a health kit icon, is for selecting health packages. This interface centralizes various health management tools, making it easy for users to navigate their health information.

Scene 5

The fifth scene focuses on providing users with a detailed report of their health. It shows specific metrics like weight and height levels, indicating whether they are at low risk. Users can switch between two views: a word view that provides a textual summary and a chart view that offers a graphical representation of their data. This dual-view system allows users to understand their health status comprehensively. A heart icon offers help, and a back button facilitates easy navigation.

Scene 6

Scene six provides a closer look at the self-monitoring feature. Users can fill in their health metrics, including weight, height, blood pressure, and blood sugar levels. This scene emphasizes the importance of regular health monitoring and provides an easy-to-use interface for logging and tracking personal health data. Five buttons correspond to different health metrics, ensuring users can accurately and consistently record their health information.

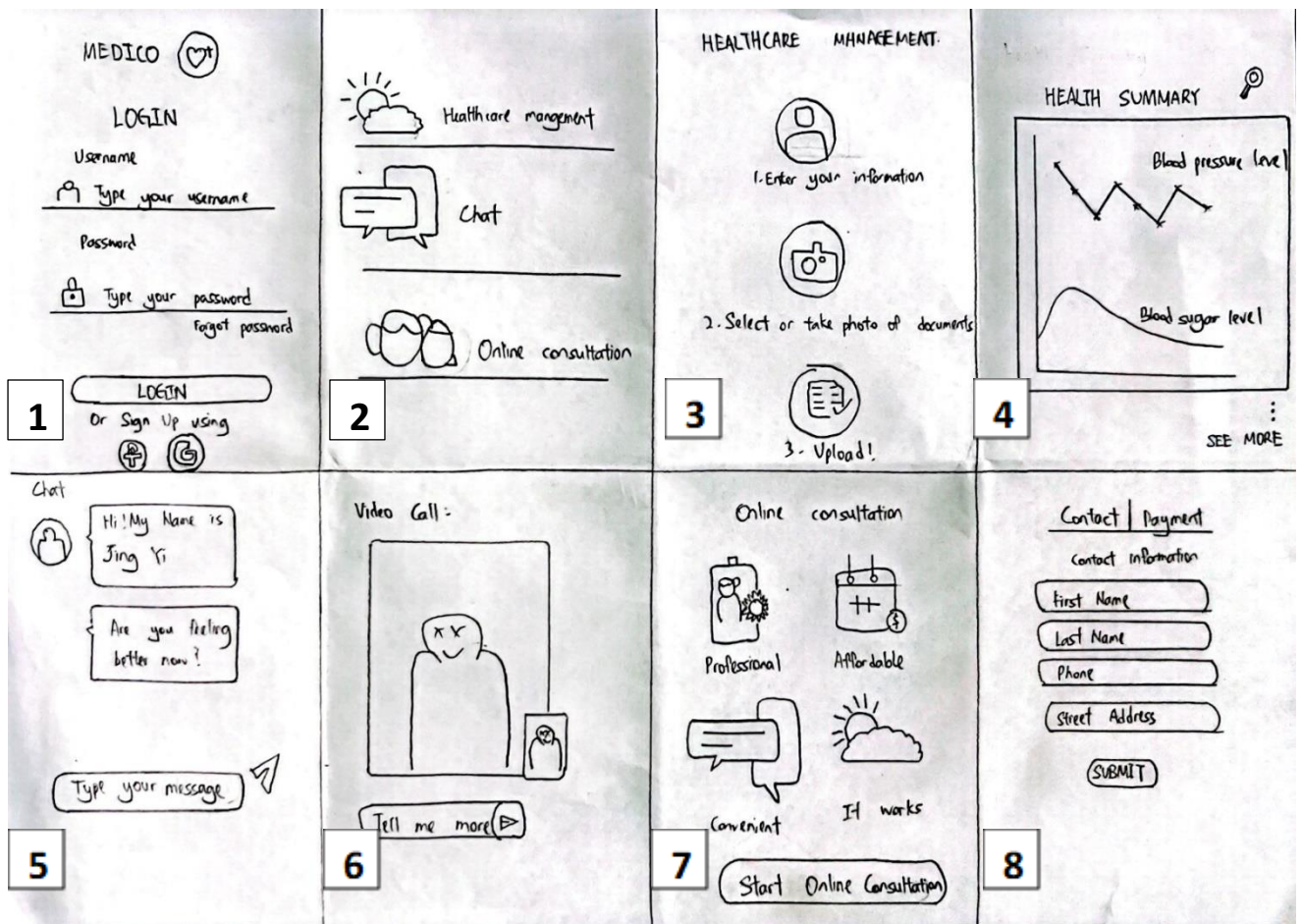
Scene 7

The seventh scene allows users to select health packages tailored to different demographics: children, men and women, and the elderly. Each package includes specific treatments and options suited to the age and needs of the user. This feature ensures that users can find appropriate healthcare solutions and book treatments conveniently through the app. The interface includes visual representations of each demographic group, making it easy for users to select the right package.

Scene 8

The final scene is dedicated to the "Online Therapy" interface. Users can choose between two therapy methods: online or face-to-face (ftf). This flexibility allows users to select the therapy mode that best suits their needs and preferences. The interface includes a "virtual teacher" icon to represent online therapy sessions. A help icon shaped like a heart is available if users need guidance, and a back button ensures they can easily return to the previous screen. Each interface includes a heart-shaped help button for user assistance and a back button for easy navigation, ensuring users can comfortably move through the app and find support when needed.

2.5 CHOH JING YI



Scene 1:

Firstly, the users open the MEDICO application and users need to log in to their account. To log in to the MEDICO account, users need to enter their email and passwords. If not registered, they can sign up using a Facebook or Google account. Otherwise, sign-up using Facebook account or Google account. In case of a forgotten password, users can click on the 'Forgot Password' link on the login screen, enter their email address, check their inbox for a reset link, follow the link, and create a new password. They can then log in with the new password.

Scene 2:

After logging in, users are presented with home page of the application which contains four main functions which will bring the user to the respective page such as: Healthcare Management, Chat, and Online Consultation. By selecting Healthcare Management, users can update personal health records, and track health summary. The Chat feature allows users to communicate with listener while the Online Consultation option provides access to virtual medical consultations with licensed doctors.

Scene 3:

This is the page of "HEALTHCARE MANAGEMENT". First, users need to input personal health information such as weight, height, blood pressure, and blood sugar levels. Besides that, users can select or take a photo of documents after they go to a medical checkup. Then, there is an "Upload" button to upload the document for analysis and save it for further use.

Scene 4:

This page displays the results of a health data summary, including blood pressure and blood sugar levels. Users can press "SEE MORE" to view an in-depth analysis of their health data, track trends over time, and receive personalized health recommendations based on their input. Additionally, the system can alert users to any abnormal readings and suggest necessary actions or medical consultations.

Scene 5:

Upon clicking the "Chat" button on the homepage, users are seamlessly directed to this page. Here, they have the option to select a listener with whom they'd like to communicate and then convey their difficulties or concerns.

Scene 6:

To convey their concerns in more detail, users can also choose to initiate a video call with the listener, allowing for a more personal and in-depth conversation about their difficulties.

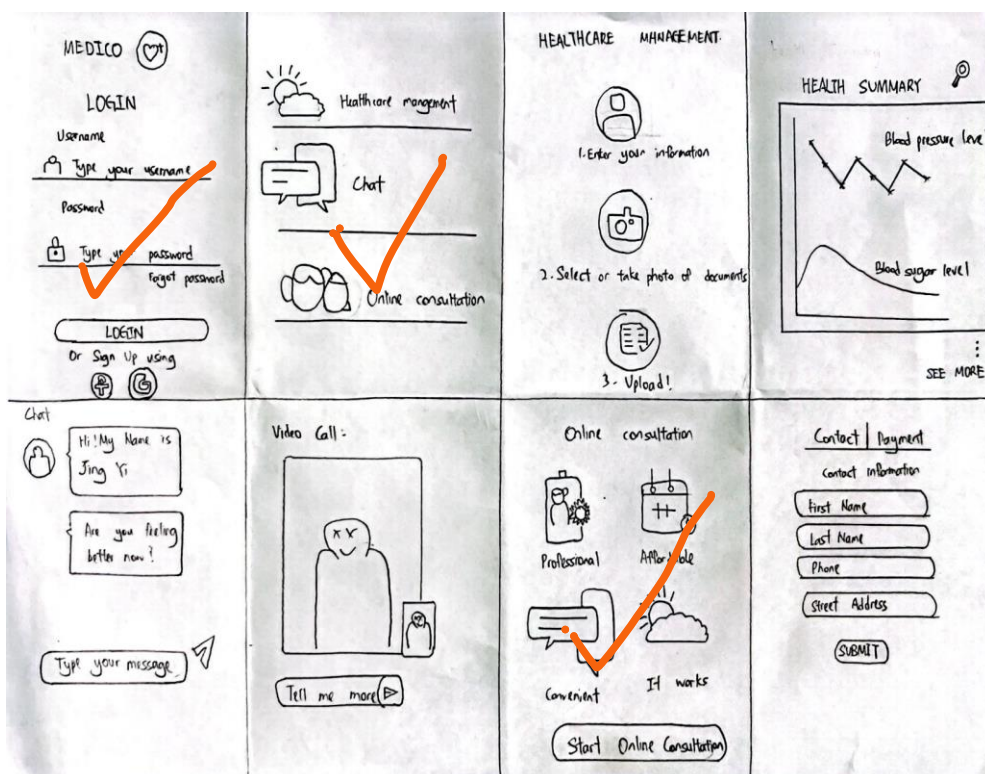
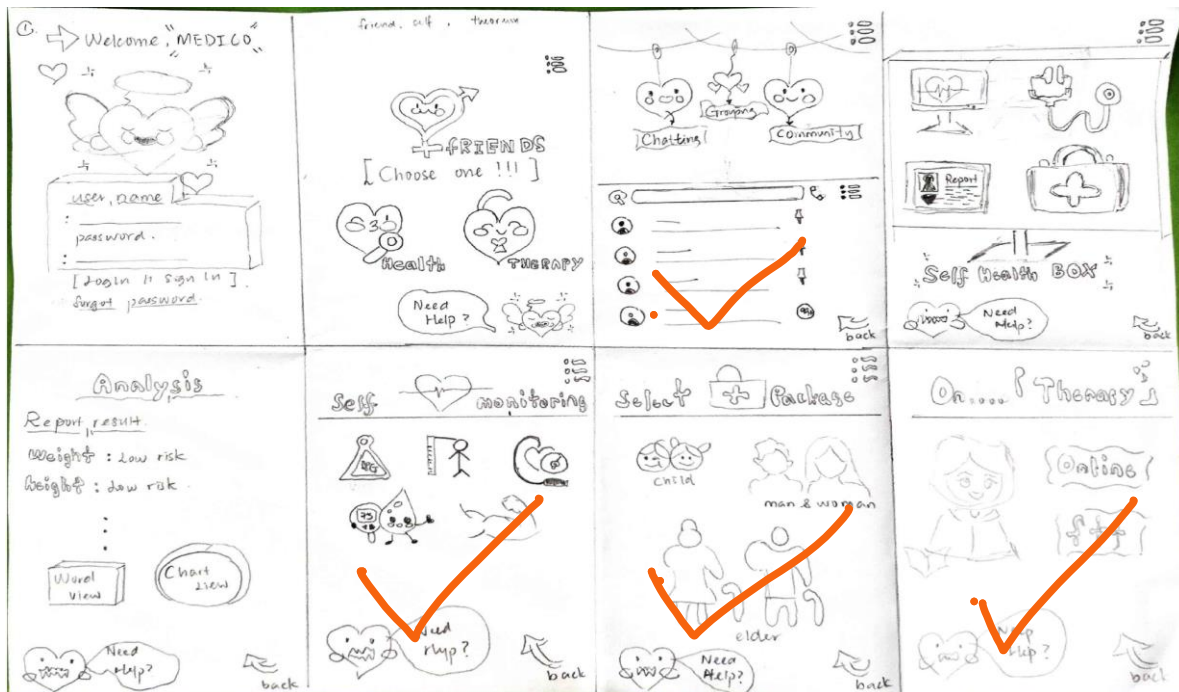
Scene 7:

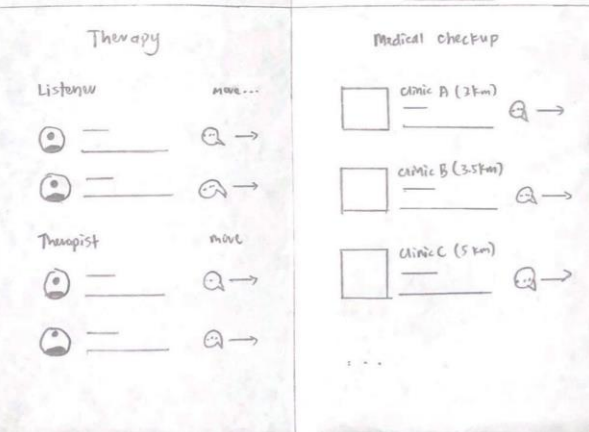
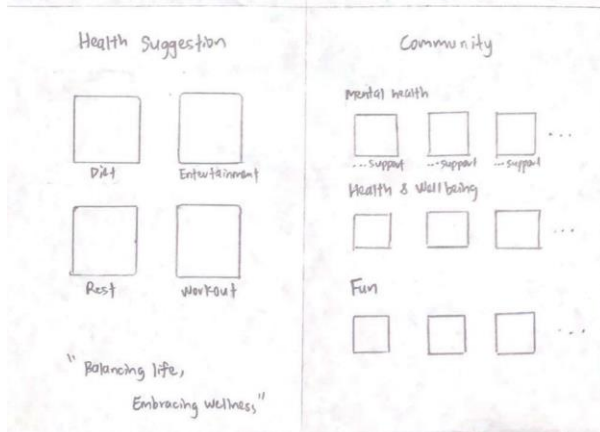
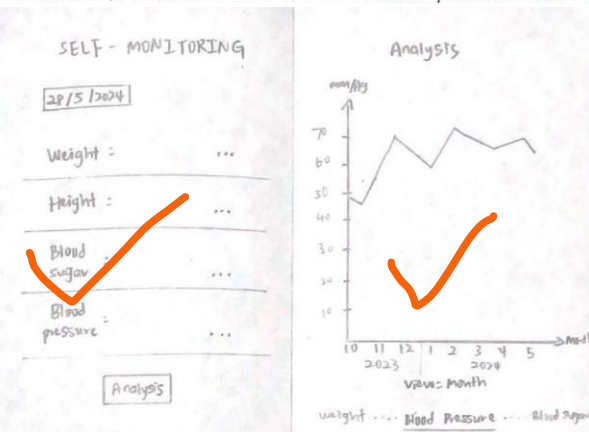
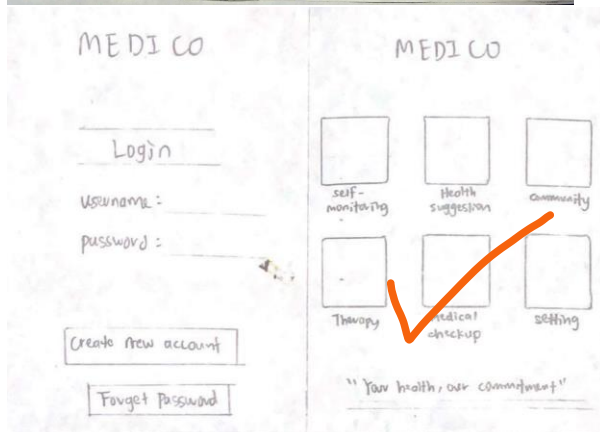
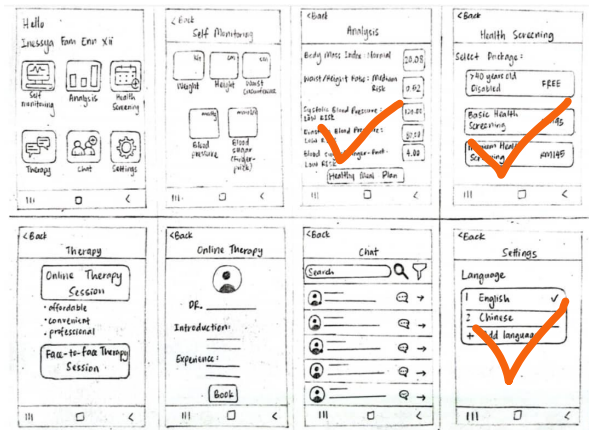
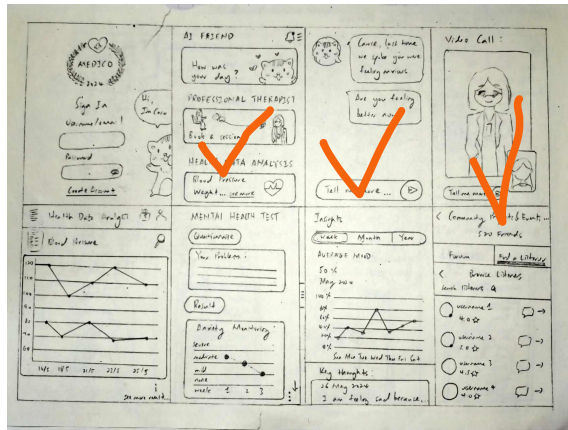
Upon clicking the "Online Consultation" button on the homepage, users are seamlessly redirected to this page. Here, the online consultation section is displayed, accompanied by a description that highlights the consultants as professional, affordable, convenient, and effective. Users can press "Start Online Consultation" to find a suitable consultant based on their description.

Scene 8:

After finding a suitable consultant, the user needs to leave their personal contact information, such as name, phone number, and street address, and click the "SUBMIT" button. Then, the user is required to make a payment to complete the appointment with the professional consultant and arrange a virtual meeting with them.

2.5 Scan of voted design layouts





1. ELIJAH SHE YU SHENG- EVERYONE

Reasons: He wants the good things of everyone's design be combined and become a better interface. Each interface has its own unique feature can be combined to form a new interface we developed.

2. CARRIE TEO HUI XIN- vote for INESSYA FAM ENN XII + CARRIE TEO HUI XIN

Reasons: Both designs are very detailed and specifically address the problems we have previously discovered and discussed. Both designs provide comprehensive solutions to the issues we discovered. The designs are neat and beginner-friendly, featuring intuitive icons accompanied by explanatory text, ensuring that users understand each icon's function.

3. INESSYA FAM ENN XII- vote for INESSYA FAM ENN XII + CARRIE TEO HUI XIN

Reasons: Both designs look neat and easy to use by beginners. This is because all icons look easy to understand and they have additional text to let users to understand what the function of all icons is.

4. LEE YIN SHEN- vote for INESSYA FAM ENN XII + LEE YIN SHEN

Reasons: Their design of each interface are straightforward and user friendly which can make the users easy to use the app. Also, the use of icons is just enough, not too fancy which is able to make the app more attractive. What's more, both functions can solve the proposed problems effectively.

5. CHOH JING YI- voted to INESSYA FAM ENN XII

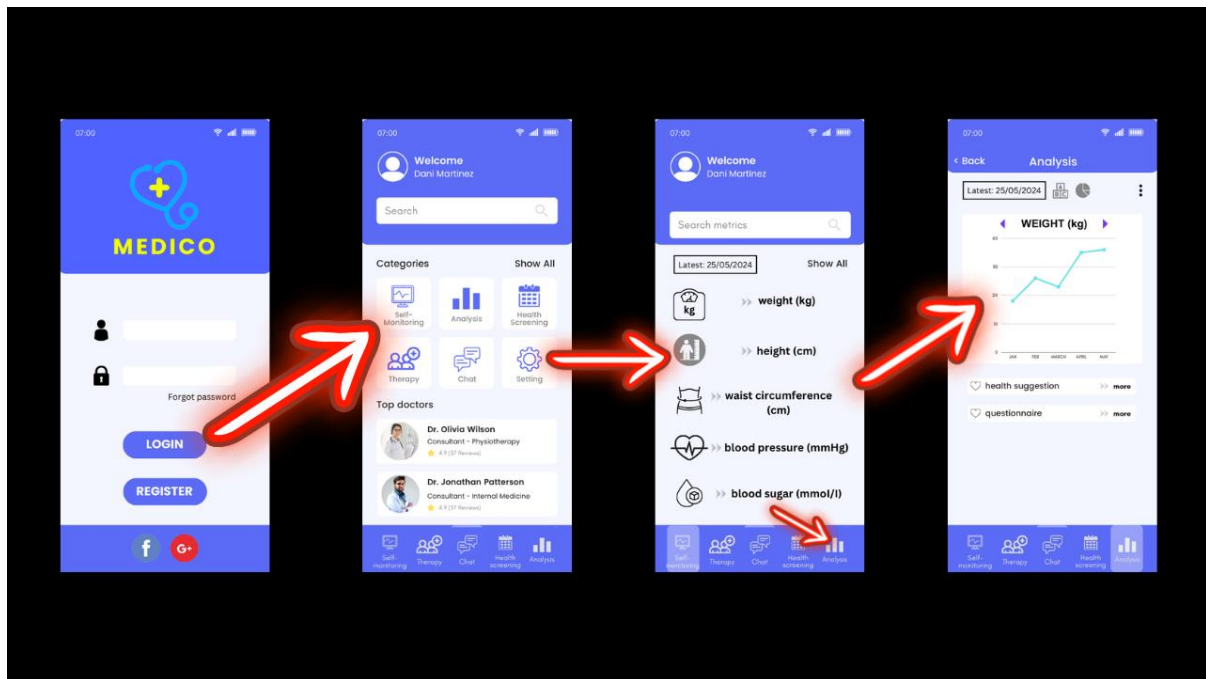
Reasons: Her design is exceptionally detailed and oriented towards the problems we identified and discussed in Project 2. It effectively addresses and resolves the problem we discovered, demonstrating an informative detail to the users.

Result

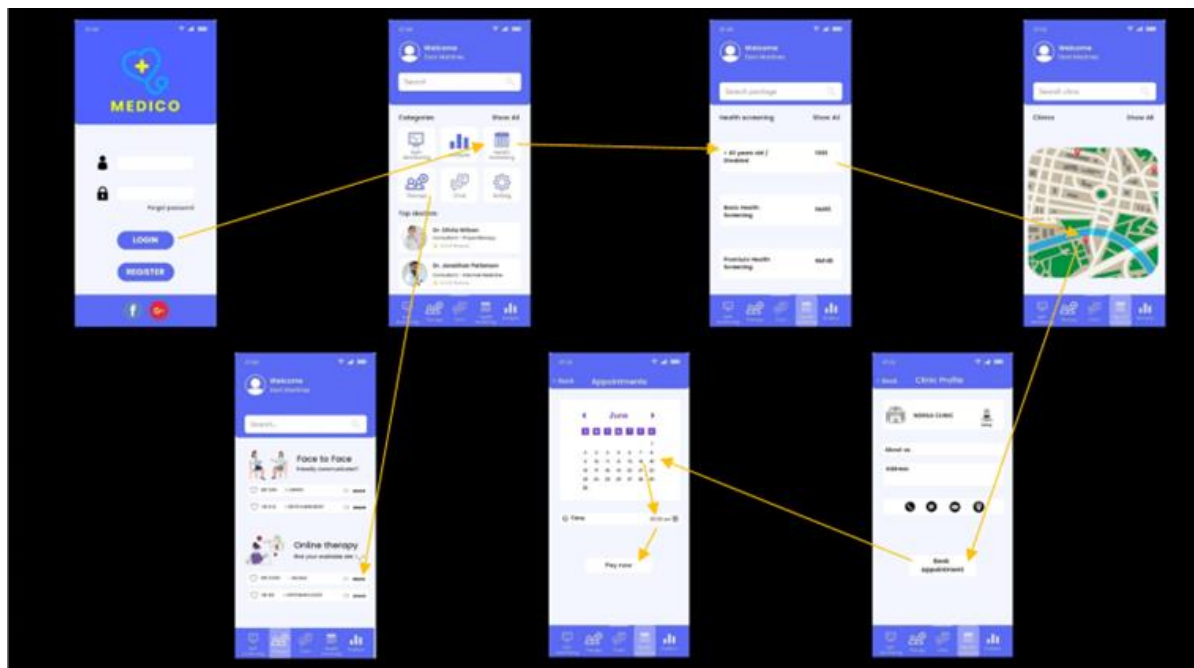
Based on the majority voting, we all decided to use INESSYA FAM ENN XII's design as a basic. Then, we refine it by combining all unique part of everyone's design to enhance our application's design.

3.0 Wireframe

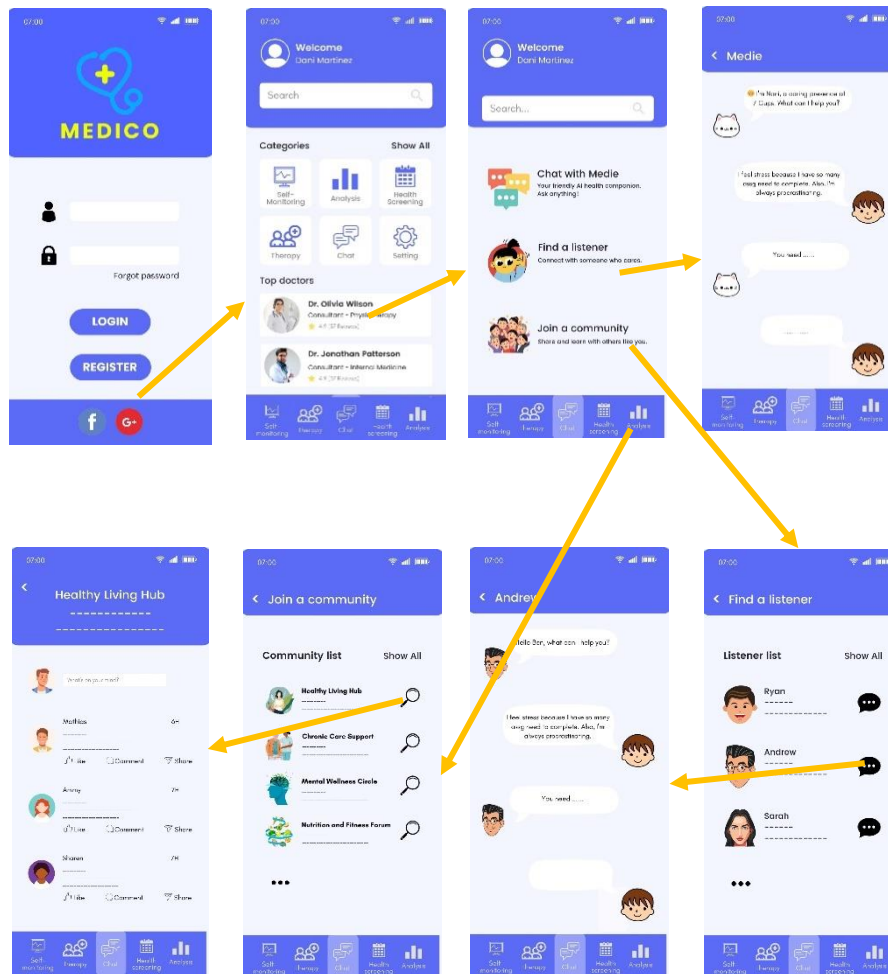
3.1 Task 1: Integrated healthcare record management system



3.2 Task 2: Provide Free Online Consultation & Reminders



3.3 Task 3: Online Chatting Platform



3.4 Justification of Design

In our system, we implemented four Gestalt principles, throughout the whole design, which are Similarity, Proximity, Continuation and Figure & Ground. We will only pick some interfaces to show the Gestalt principles.

Under Law of Similarity, each icon on our home page after the login page has different images. However, these icons are similar in shapes and style. Therefore, they are perceived to be one group. Users will know exactly what UI elements to look for.

Additionally, under Law of Proximity, in our home page, the icons below categories are close together and the top doctors are put close enough to be seen as one group. Grouping related elements are critical to give the interface a clear visual hierarchy.

Under Law of Continuation, elements in self-monitoring page are arranged in a horizontal linear arrangement. Hence, they are perceived to be related. Our elements in self-monitoring screen are arranged horizontally and in health screening screen it is arranged vertically communicate relatedness.

Under Law of Figure & Ground, this ability allows users to determine what they should pay attention. In our clinics page after the health screening page, we used colour, shading, highlights in our map to cue the user in on what has focus and ground depth. We also placed buttons under every page that look like press-able buttons, where users will know the icons are linked to their respective pages.

Moreover, we also followed Shneiderman's Golden Rules to create user-friendly interfaces. In every page after the home page, there will be icons at the bottom of the page to enable frequent users to use shortcuts. Experienced users should be able to speed up their interactions by using the shortcuts, because the icons link them back to the page of the icons they clicked.

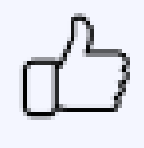
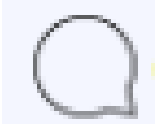
Furthermore, we also permit easy reversal of actions in every page. There is a back button at the top left of every page where users should be able to undo actions easily if they make a mistake. Finally, we aim to reduce short-term memory load. We minimize the user's need to remember information from one part of the interface to another. For example, we use visual aids like menus, icons, and labels to reduce cognitive load.

4.0 Description and justification of metaphors

Metaphor	Name	Function
	User	To show the user's avatar.
	Search	Enable user to search for the functions they want.
	Self-monitoring	Enable user to input their personal health data inside this function.
	Analysis	Enable user to view the analysis of their personal data inside this function.
	Health screening	Enable user to book medical checkup appointment inside this function.
	Therapy	Enable user to find therapist to get mental assistance.
	Chat	Enable user to chat with AI, find a listener and join a community.

	Setting	Enable user to customize the app like change the theme colour and language.
	Weight	Enable user to enter their weight.
	Height	Enable user to enter their height.
	Waist circumference	Enable user to enter their waist circumference.
	Blood pressure	Enable user to enter their blood pressure level.
	Blood sugar	Enable user to enter their blood sugar level.
Latest: 25/05/2024	Date	Enable user to view their previous health data or input their previous health data.

	Face-to-face therapy	Enable user to choose face-to-face therapy service.
	Online therapy	Enable user to choose online therapy service.
	AI chatting bot	Enable user to chat with AI chatting bot.
	Listener	Enable user to find a listener.
	Community	Enable user to join a community.
	Chat	Enable user to chat with listener.
	Learn more	Enable user to see more detail of a community.

	Like	Enable user to like other's post in a community.
	Comment	Enable user to comment other's post in a community.
	Share	Enable user to share other's post in a community.